BASAVESHWAR ENGINEERING COLLEGE, BAGALKOTE

Department of Computer Science and Engineering

Project Interaction Sheet

Project	AI-Based Warehouse Automation System Using Reinforcement Learning for Intelligent Robot Navigation
Project ID / Batch	07 / 2026–27
Team	Goutam A Iti (2BA23CS035), Prerana C Patil (2BA23CS073), Srushti Bingikoppad (2BA23CS109)
Project guide	Prof. Jyoti M. Hurakadli
Date	30 October 2026
Overall completion	**93%** — 50 of 54 planned items completed as of Demo 2

Status by phase

Week	Phase	Completed	Status	Yet to be completed
1	Research and setup	4 / 4	100%	—
2	Warehouse simulator	5 / 5	100%	—
3	RL environment	5 / 5	100%	—
4	RL training (PPO)	3 / 4	75%	Q-learning / DQN validation
5	Tasks and task queue	4 / 4	100%	—
6	Advanced navigation (obstacles, battery)	5 / 5	100%	—
7	Evaluation	7 / 8	88%	Dijkstra baseline comparison
8	Web dashboard	6 / 6	100%	—
9	Integration and testing	6 / 6	100%	—
10	Documentation and demo	5 / 7	71%	Final project report; architecture diagrams for the report
	Total	**50 / 54**	**93%**	**4 items**

Work completed (till Demo 2)

- Warehouse simulator with six scenarios: shelves, pick points, delivery stations, chargers, moving obstacles and battery
- Gymnasium RL environment and six trained PPO policies, each retrained with three seeds
- A* and BFS baselines; paired evaluation on 30 identical episodes per controller
- Results: PPO equals optimal A* on static floors (30/30), is faster but collides in traffic, and learns the battery task after a reward fix (0/30 → 27/30)
- Reward-weight ablation and battery reliability study (best checkpoint: 82 of 90 episodes)
- FastAPI backend and four-tab web portal; multi-robot Live floor with task queue and urgent priority
- 276 automated tests; all 540 published episodes reproduce exactly

Yet to be completed (for the final demo)

☐ Q-learning / DQN validation run
☐ Dijkstra baseline comparison
☐ Final project report
☐ Architecture and workflow diagrams for the report

Guide's remarks

Signatures

Role	Name	Signature	Date
Student	Goutam A Iti		
Student	Prerana C Patil		
Student	Srushti Bingikoppad		
Project guide	Prof. Jyoti M. Hurakadli		